ASCE 41-23 Tier 1 Checklists

FIRM:	
PROJECT NAME:	
SEISMICITY LEVEL:	
PROJECT NUMBER:	
COMPLETED BY:	
DATE COMPLETED:	
REVIEWED BY:	
REVIEW DATE:	

Legend: C = Compliant, NC = Noncompliant, N/A = Not Applicable, U = Unknown

© 2023 American Society of Civil Engineers

Project Name _____
Project Number _____

17.18CP Structural Checklist for Building Types URM: Unreinforced Masonry Bearing Walls with Flexible Diaphragms and URMa: Unreinforced Masonry Bearing Walls with Stiff Diaphragms

Table 17-36. Collapse Prevention Structural Checklist for Building Types URM and URMa.

Status				Evaluation Statement	Tier 2 Reference	Commentary Reference	Comments
Low and Moderate Seismicity							
Seismic-Force-Resisting System							
C ☐	**NC** ☐	**N/A** ☐	**U** ☐	REDUNDANCY: The number of lines of shear walls in each principal direction is greater than or equal to 2.	5.5.1.1	A.3.2.1.1	
C ☐	**NC** ☐	**N/A** ☐	**U** ☐	SHEAR STRESS CHECK: The shear stress in the unreinforced masonry shear walls, calculated using the Quick Check procedure of Section 4.4.3.3, is less than 30 lb/in.^{2} (0.21 MPa) for clay units and 70 lb/in.^{2} (0.48 MPa) for concrete units.	5.5.3.1.1	A.3.2.5.1	
Connections							
C ☐	**NC** ☐	**N/A** ☐	**U** ☐	WALL ANCHORAGE: Exterior concrete or masonry walls that are dependent on the diaphragm for lateral support are anchored for out-of-plane forces at each diaphragm level with steel anchors, reinforcing dowels, or straps that are developed into the diaphragm. Connections have strength to resist the connection force calculated in the Quick Check procedure of Section 4.4.3.7.	5.7.1.1	A.5.1.1	
C ☐	**NC** ☐	**N/A** ☐	**U** ☐	WOOD LEDGERS: The connection between the wall panels and the diaphragm does not induce cross-grain bending or tension in the wood ledgers or top plates fastened to the walls.	5.7.1.3	A.5.1.2	
C ☐	**NC** ☐	**N/A** ☐	**U** ☐	TRANSFER TO SHEAR WALLS: Diaphragms are connected for transfer of seismic forces to the shear walls.	5.7.2	A.5.2.1	
C ☐	**NC** ☐	**N/A** ☐	**U** ☐	GIRDER–COLUMN CONNECTION: There is a positive connection using plates, connection hardware, or straps between the girder and the column support.	5.7.4.1	A.5.4.1	

Legend: C = Compliant, NC = Noncompliant, N/A = Not Applicable, U = Unknown

Status				Evaluation Statement	Tier 2 Reference	Commentary Reference	Comments
High Seismicity (Complete the Following Items in Addition to the Items for Low and Moderate Seismicity)							
Seismic-Force-Resisting System							
C ☐	**NC** ☐	**N/A** ☐	**U** ☐	PROPORTIONS: The height-to-thickness ratio of the shear walls at each story is less than the following: Top story of multistory building, 9; First story of multistory building, 15; and All other conditions, 13.	5.5.3.1.2	A.3.2.5.2	
C ☐	**NC** ☐	**N/A** ☐	**U** ☐	MASONRY LAYUP: Filled collar joints of multiwythe masonry walls have negligible voids.	5.5.3.4.1	A.3.2.5.3	
Diaphragms (Stiff or Flexible)							
C ☐	**NC** ☐	**N/A** ☐	**U** ☐	DIAPHRAGM CONTINUITY: Floor and roof diaphragms do not have expansion joints or vertical offsets, such as split levels, sawtooth, or clerestory configurations.	5.6.1.1	A.4.1.1	
C ☐	**NC** ☐	**N/A** ☐	**U** ☐	ROOF CHORD CONTINUITY: All chord elements are continuous, regardless of changes in roof elevation.	5.6.1.1	A.4.1.3	
C ☐	**NC** ☐	**N/A** ☐	**U** ☐	OPENINGS AT SHEAR WALLS: Diaphragm openings immediately adjacent to the shear walls are less than 25% of the wall length.	5.6.1.3	A.4.1.4	
C ☐	**NC** ☐	**N/A** ☐	**U** ☐	OPENINGS AT EXTERIOR MASONRY SHEAR WALLS: Diaphragm openings immediately adjacent to exterior masonry shear walls are not greater than 8 ft (2.4 m) long.	5.6.1.3	A.4.1.6	
Flexible Diaphragms							
C ☐	**NC** ☐	**N/A** ☐	**U** ☐	CROSSTIES: There are continuous crossties between diaphragm chords to distribute the out-of-plane wall anchorage forces into the diaphragm. Where each out-of-plane connection does not have a continuous crosstie across the entire diaphragm, these connections are developed into subdiaphragms between crossties with a maximum length-to-width ratio of 3-to-1.	5.6.1.2	A.4.1.2	

Legend: C = Compliant, NC = Noncompliant, N/A = Not Applicable, U = Unknown

© 2023 American Society of Civil Engineers

C	NC	N/A	U			
☐	☐	☐	☐	STRAIGHT SHEATHING: All straight-sheathed diaphragms have horizontal spans less than 24 ft (7.3 m) and aspect ratios less than 2-to-1 in the direction being considered.	5.6.2	A.4.2.1
C	NC	N/A	U			
☐	☐	☐	☐	DIAGONALLY SHEATHED AND UNBLOCKED DIAPHRAGMS: All diagonally sheathed or unblocked wood structural panel diaphragms have horizontal spans less than 40 ft (12.2 m) and aspect ratios less than or equal to 4-to-1.	5.6.2	A.4.2.2
C	NC	N/A	U			
☐	☐	☐	☐	BLOCKED DIAPHRAGMS: All blocked wood structural panel diaphragms have horizontal spans less than 120 ft (36.5 m) and have aspect ratios less than or equal to 4-to-1.	5.6.2	A.4.2.3
C	NC	N/A	U			
☐	☐	☐	☐	CANTILEVERED WOOD DIAPHRAGMS: All cantilevered wood diaphragms that provide lateral support for concrete or masonry walls consist of wood structural panels and have a maximum cantilever length of 20 ft (6.1 m) if unblocked or 35 ft (10.7 m) if blocked, and a maximum ratio of cantilever length to diaphragm width of 1:2 if unblocked and 1:1 if blocked. In addition, the cantilevered diaphragm has a back-span length equal to or greater than the cantilevered portion.	5.6.2	A.4.2.4
C	NC	N/A	U			
☐	☐	☐	☐	NON-CONCRETE-FILLED DIAPHRAGMS: Bare steel deck diaphragms or steel deck diaphragms with fill other than reinforced structural concrete consist of horizontal spans of less than 120 ft (36.5 m) and have aspect ratios less than 4-to-1.	5.6.3	A.4.3.1
C	NC	N/A	U			
☐	☐	☐	☐	OTHER DIAPHRAGMS: The diaphragms do not consist of a system other than wood, steel deck, concrete, or horizontal bracing.	5.6.5	A.4.7.1

Legend: C = Compliant, NC = Noncompliant, N/A = Not Applicable, U = Unknown

Connections						
C	**NC**	**N/A**	**U**	STIFFNESS OF WALL ANCHORS: Anchors of concrete or masonry walls to wood structural elements are installed taut and are stiff enough to limit the relative movement between the wall and the diaphragm to no greater than 1/8 in. (3.1 mm) before engagement of the anchors.	5.7.1.2	A.5.1.4
☐	☐	☐	☐			
C	**NC**	**N/A**	**U**	BEAM, GIRDER, AND TRUSS SUPPORTS: Beams, girders, and trusses supported by unreinforced masonry walls or pilasters have independent secondary columns for support of vertical loads.	5.7.4.4	A.5.4.5
☐	☐	☐	☐			

Legend: C = Compliant, NC = Noncompliant, N/A = Not Applicable, U = Unknown